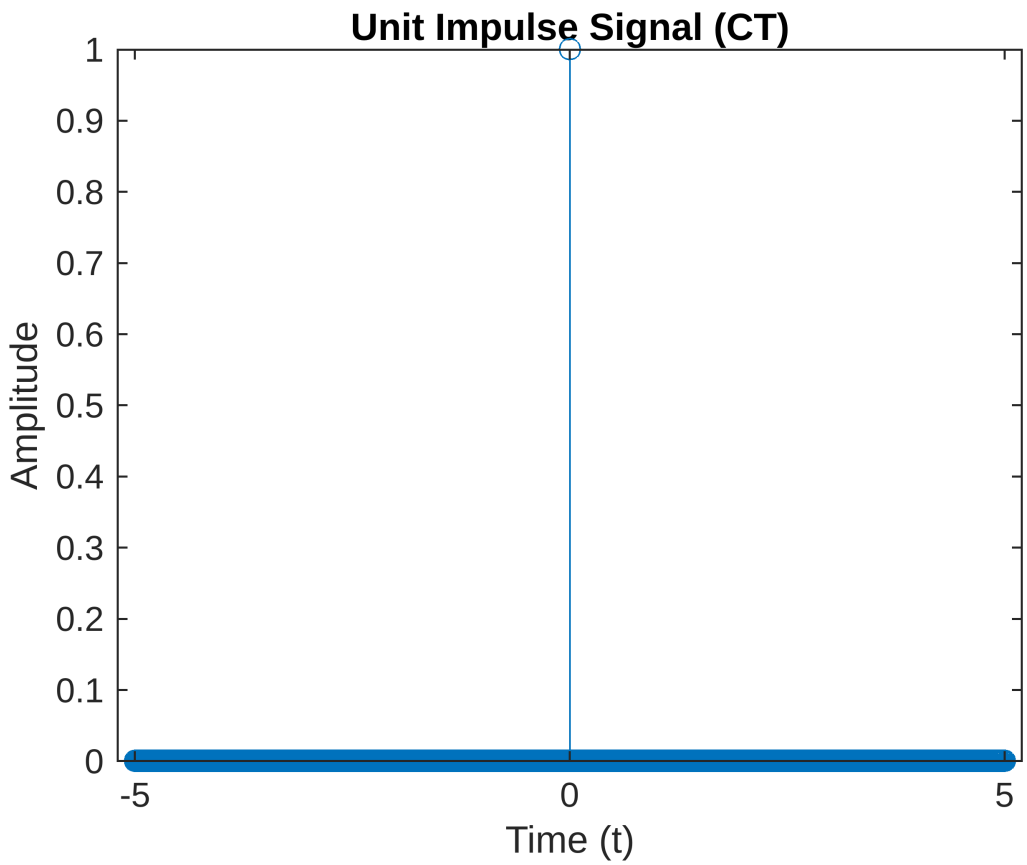
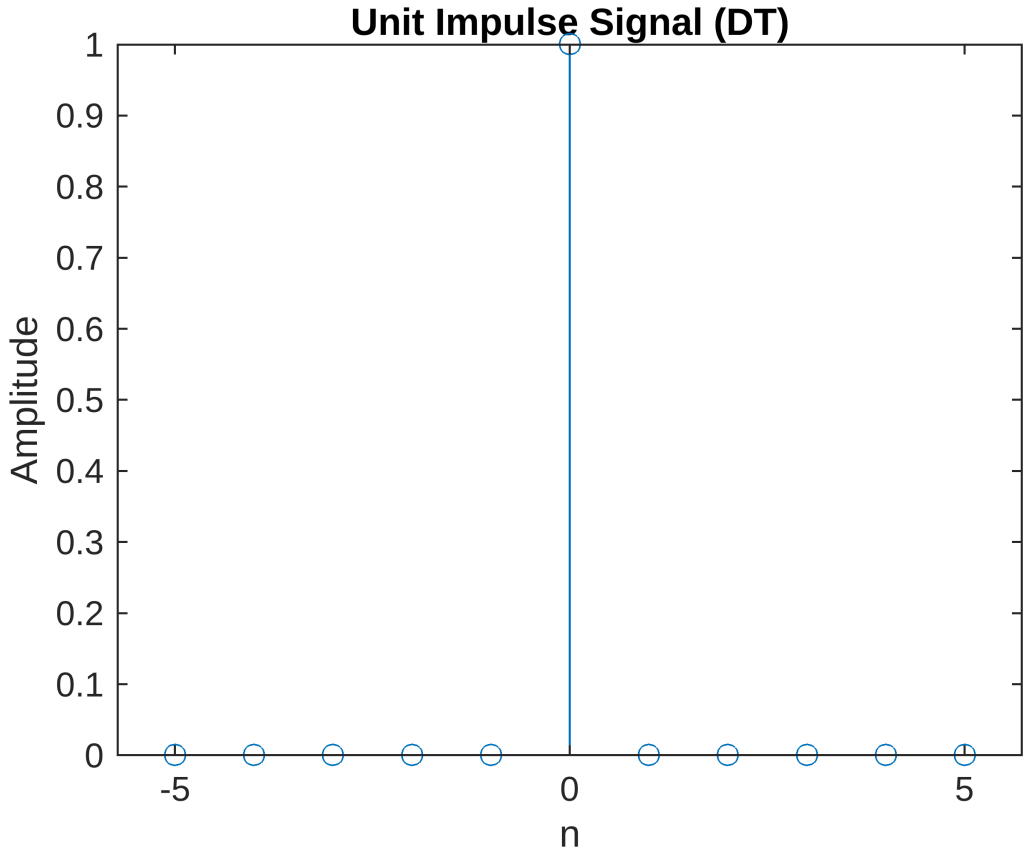
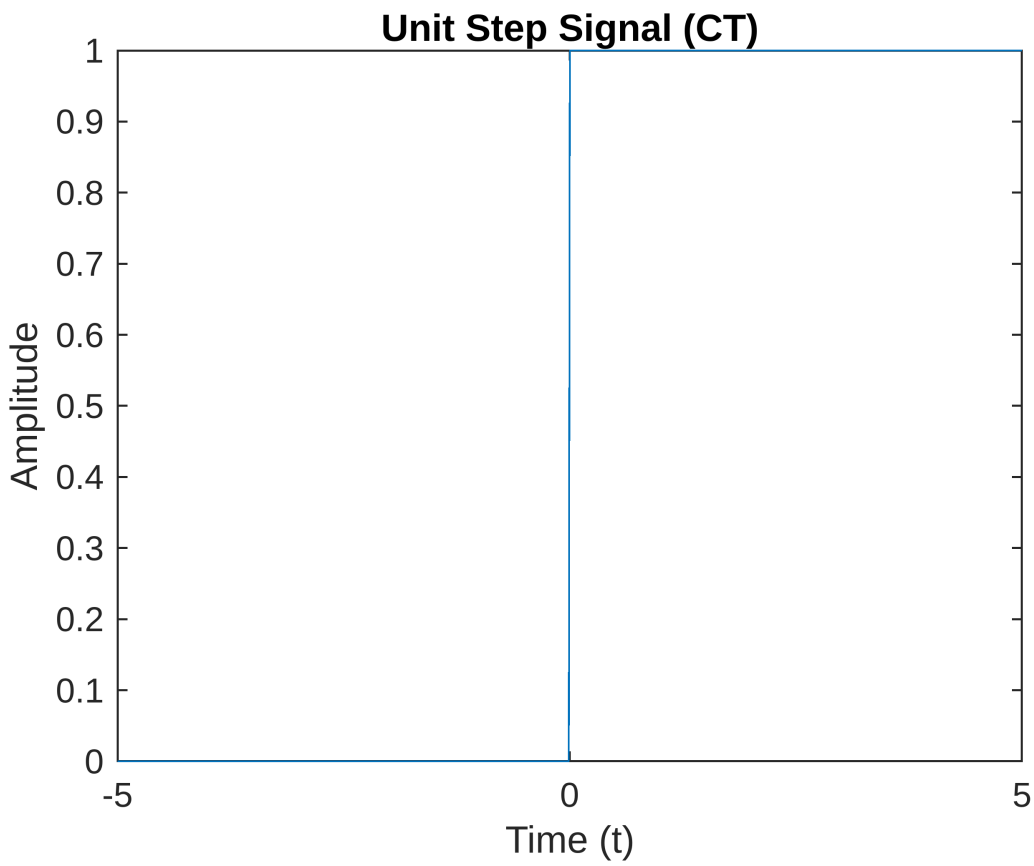
```
% 1. Program for the generation of unit impulse signal.
% Continuous-Time (CT):
t = -5:0.01:5;    % Time vector
impulse_ct = t == 0; % Unit impulse in CT
figure;
stem(t, impulse_ct);
title('Unit Impulse Signal (CT)');
xlabel('Time (t)');
ylabel('Amplitude');
```



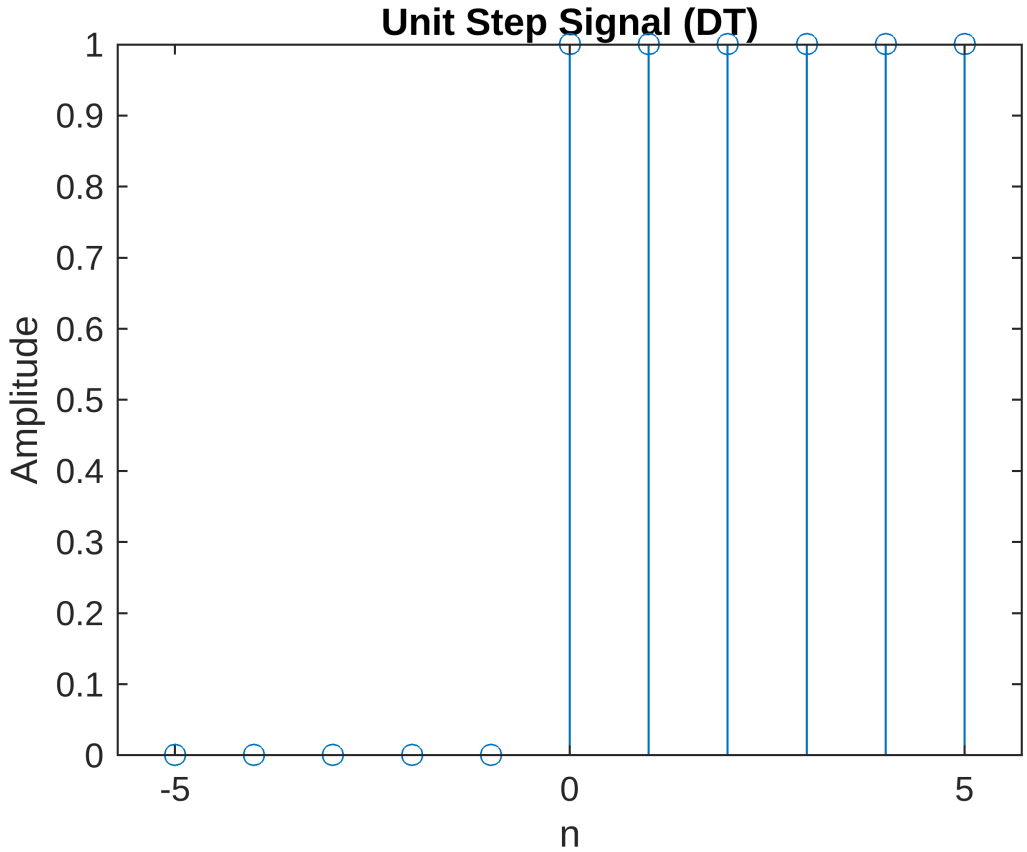

```
% Discrete-Time (DT):
n = -5:5;    % Discrete time vector
impulse_dt = (n == 0);   % Unit impulse in DT
figure;
stem(n, impulse_dt);
title('Unit Impulse Signal (DT)');
xlabel('n');
ylabel('Amplitude');
```



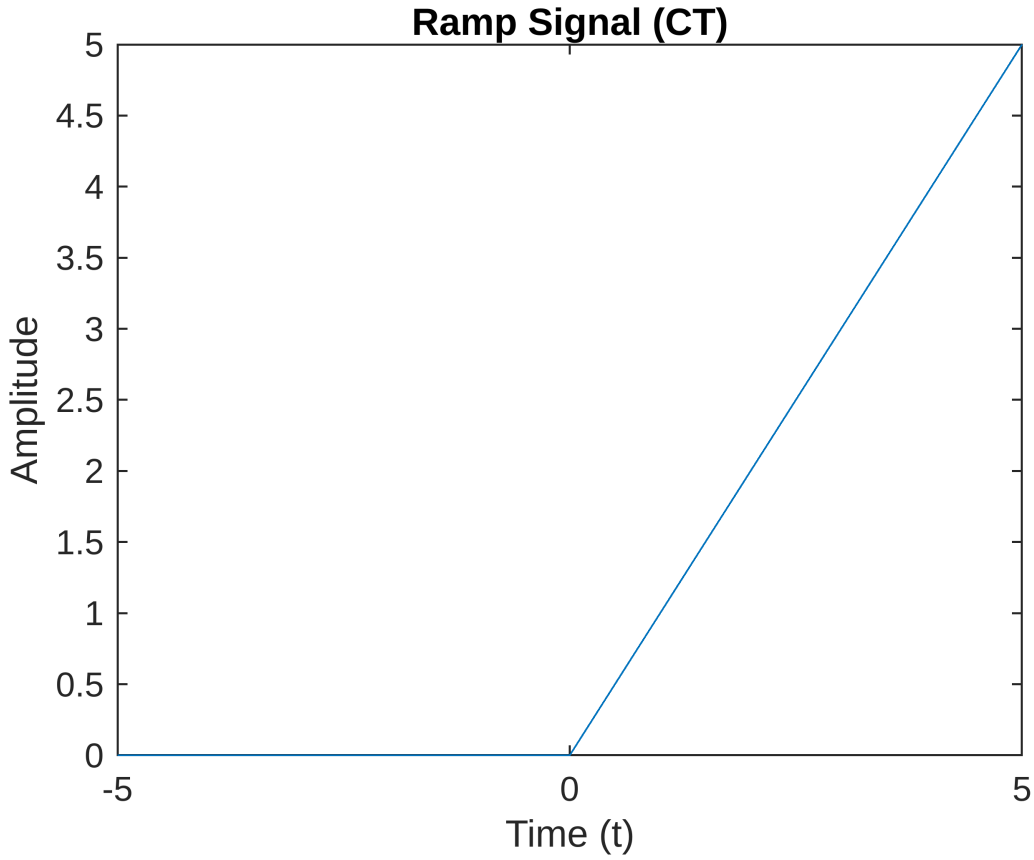

```
% 2. Program for the generation of unit step signal
% Continuous-Time (CT):
t = -5:0.01:5;    % Time vector
step_ct = t >= 0; % Unit step in CT
figure;
plot(t, step_ct);
title('Unit Step Signal (CT)');
xlabel('Time (t)');
ylabel('Amplitude');
```



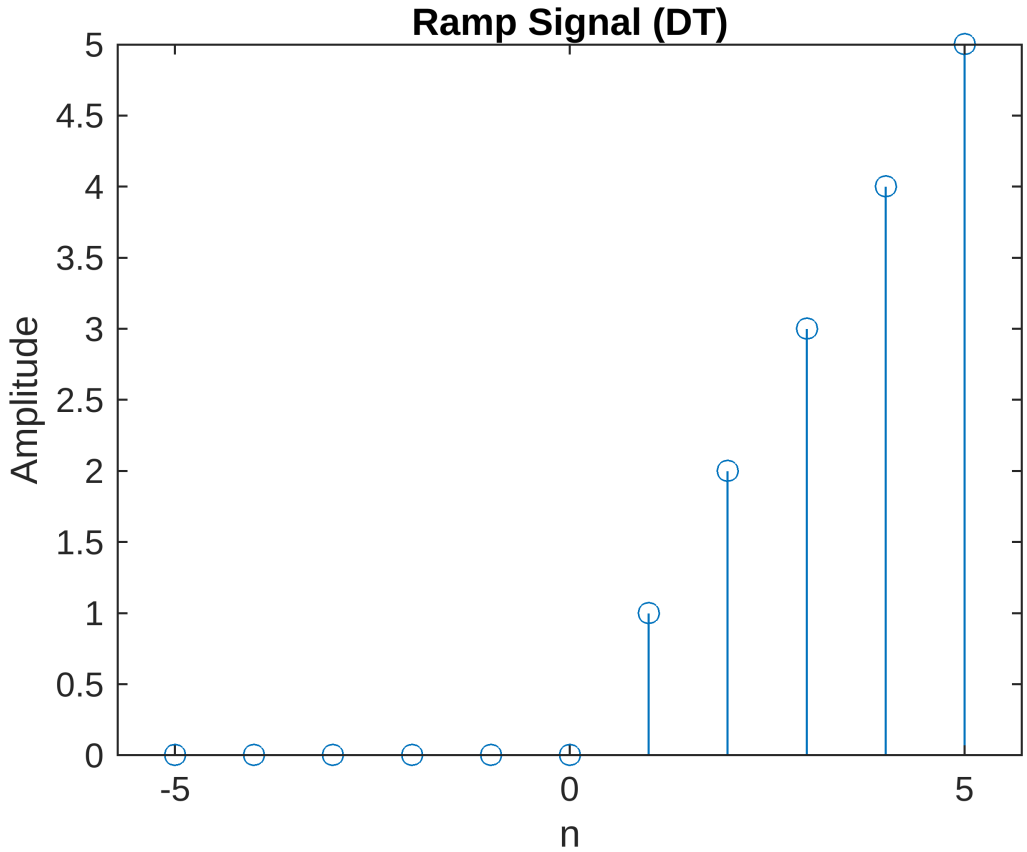

```
% Discrete-Time (DT):
n = -5:5;    % Discrete time vector
step_dt = n >= 0;   % Unit step in DT
figure;
stem(n, step_dt);
title('Unit Step Signal (DT)');
xlabel('n');
ylabel('Amplitude');
```



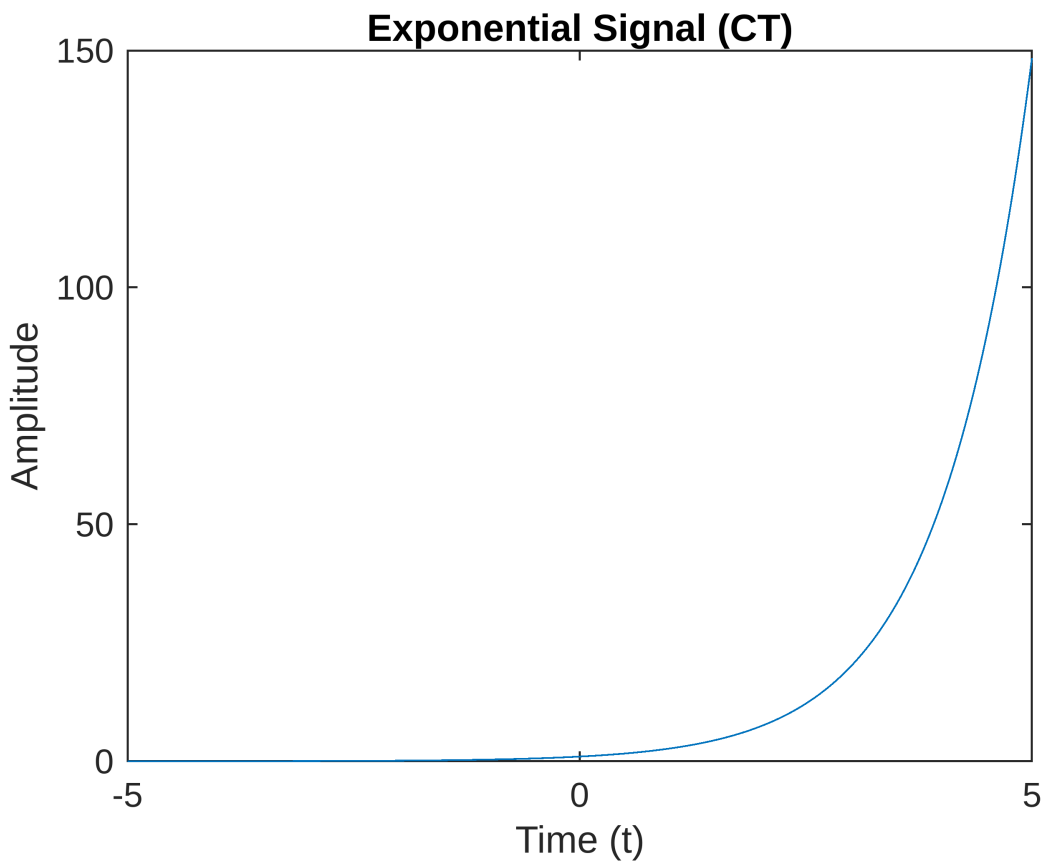

```
% 3. Program for the generation of ramp signal
% Continuous-Time (CT):
t = -5:0.01:5;    % Time vector
ramp_ct = t .* (t >= 0); % Ramp in CT
figure;
plot(t, ramp_ct);
title('Ramp Signal (CT)');
xlabel('Time (t)');
ylabel('Amplitude');
```



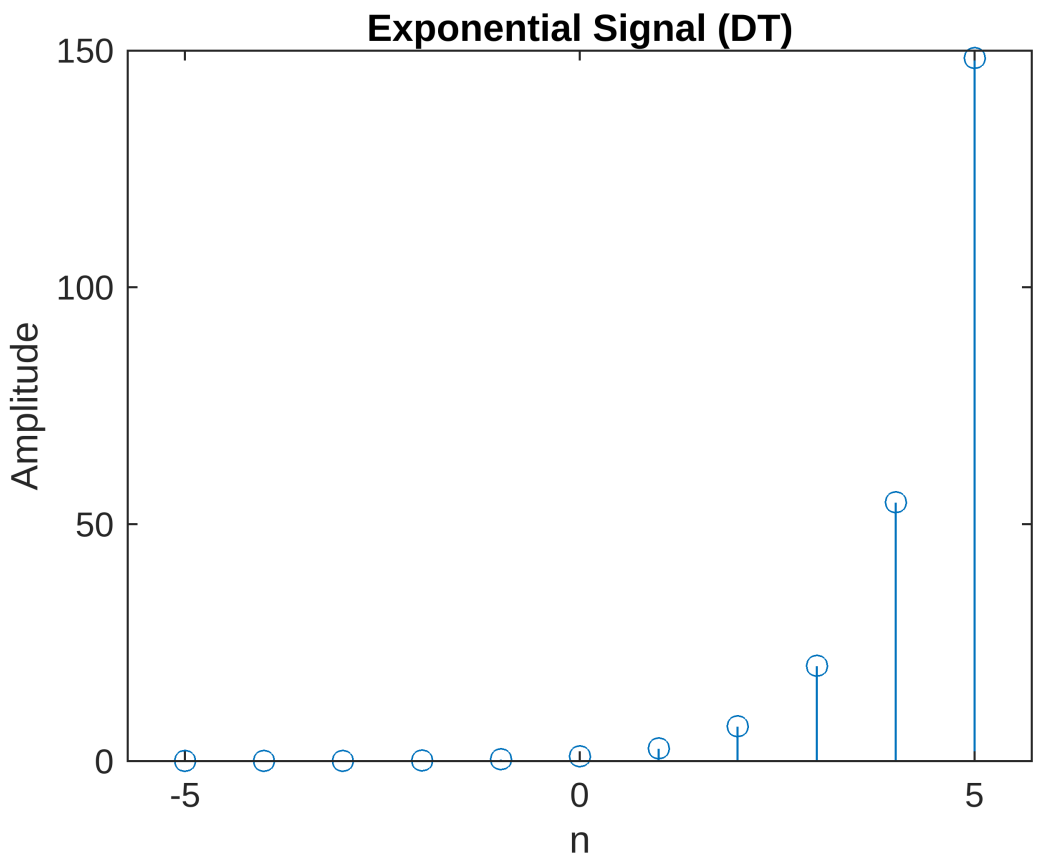

```
% Discrete-Time (DT):
n = -5:5;    % Discrete time vector
ramp_dt = n .* (n >= 0);   % Ramp in DT
figure;
stem(n, ramp_dt);
title('Ramp Signal (DT)');
xlabel('n');
ylabel('Amplitude');
```



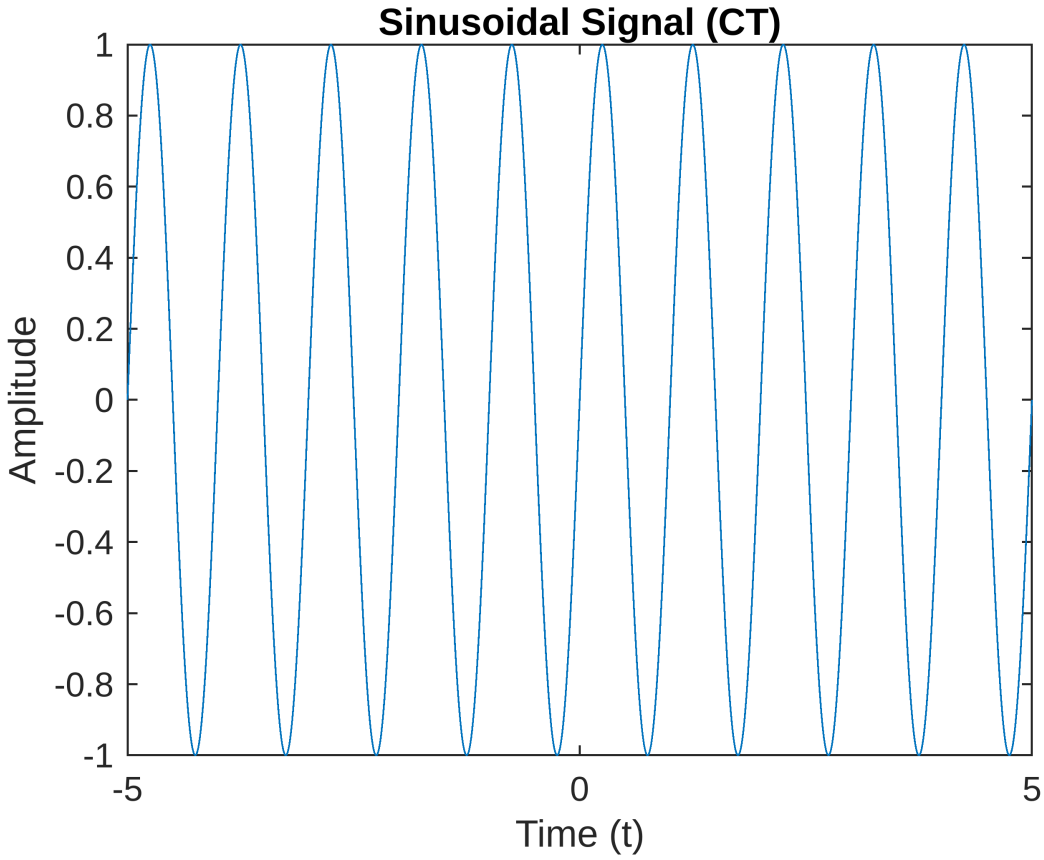

```
% 4. Program for the generation of exponential sequence
% Continuous-Time (CT):
t = -5:0.01:5;    % Time vector
exp_ct = exp(t);  % Exponential in CT
figure;
plot(t, exp_ct);
title('Exponential Signal (CT)');
xlabel('Time (t)');
ylabel('Amplitude');
```



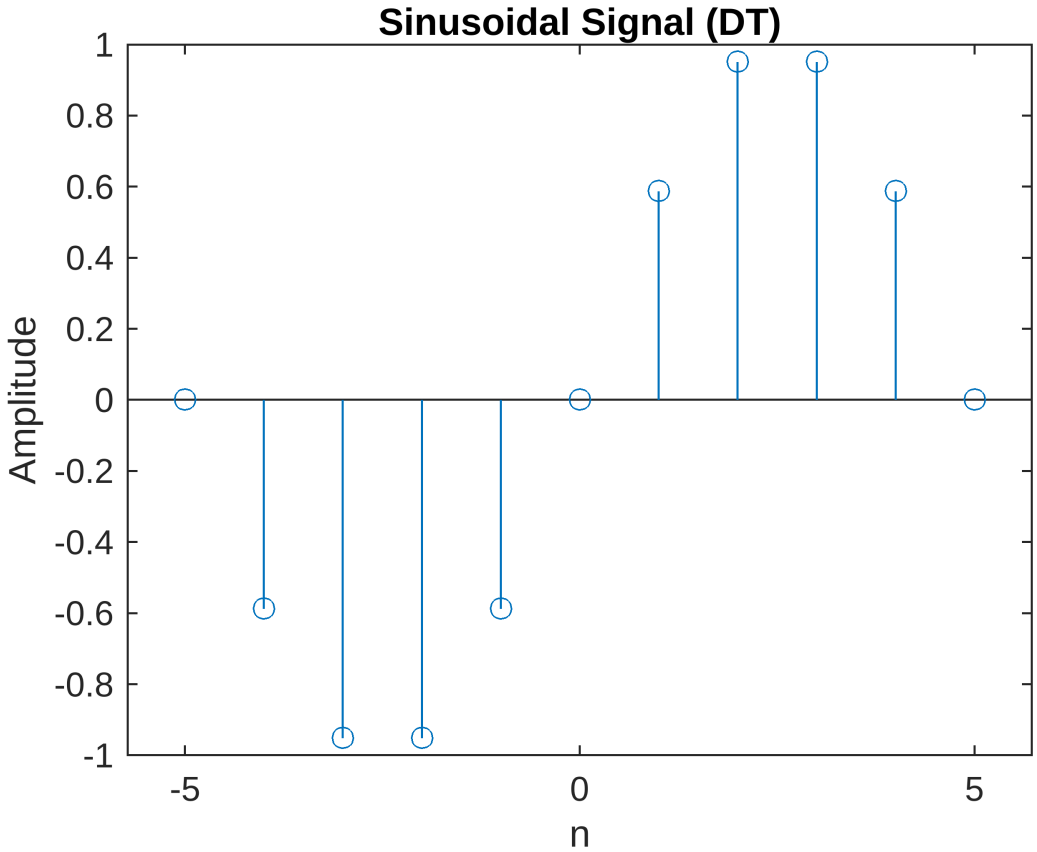

```
% Discrete-Time (DT):
n = -5:5;    % Discrete time vector
exp_dt = exp(n);   % Exponential in DT
figure;
stem(n, exp_dt);
title('Exponential Signal (DT)');
xlabel('n');
ylabel('Amplitude');
```



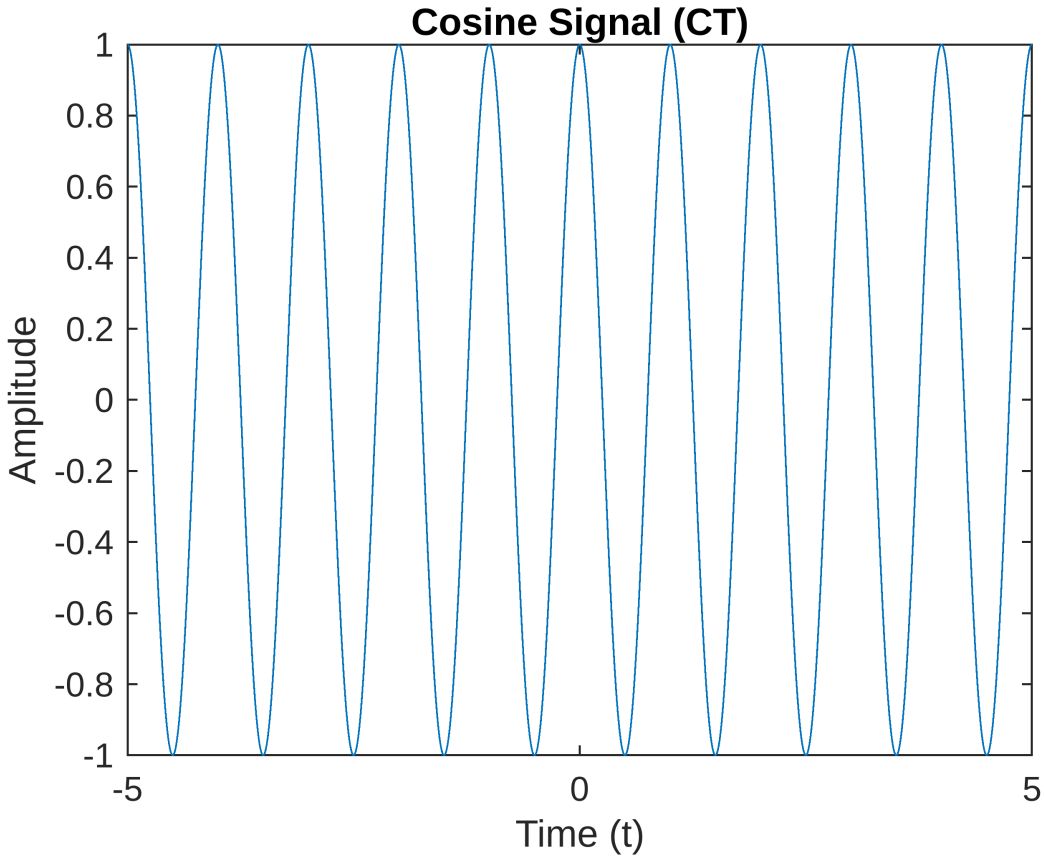

```
% 5. Program for the generation of sine signal
% Continuous-Time (CT):
t = -5:0.01:5;    % Time vector
sin_ct = sin(2*pi*t); % Sinusoidal in CT
figure;
plot(t, sin_ct);
title('Sinusoidal Signal (CT)');
xlabel('Time (t)');
ylabel('Amplitude');
```



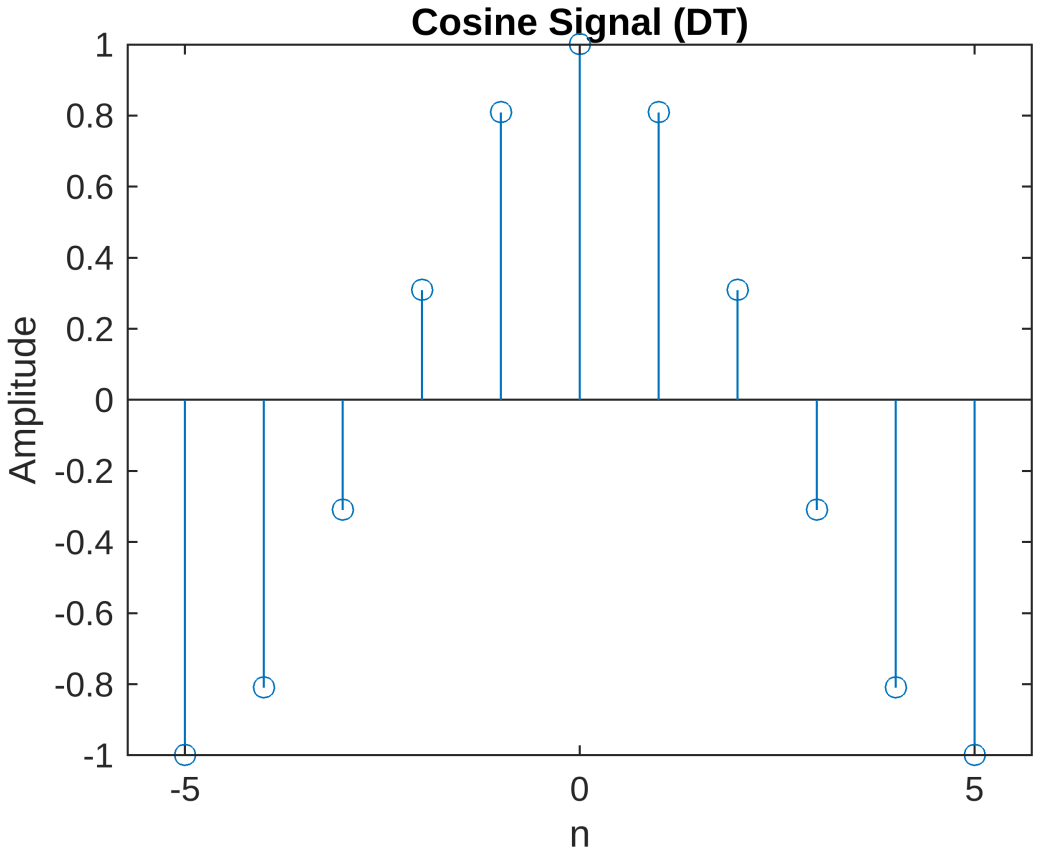

```
% Discrete-Time (DT):
n = -5:5;    % Discrete time vector
sin_dt = sin(2*pi*n/10);   % Sinusoidal in DT
figure;
stem(n, sin_dt);
title('Sinusoidal Signal (DT)');
xlabel('n');
ylabel('Amplitude');
```



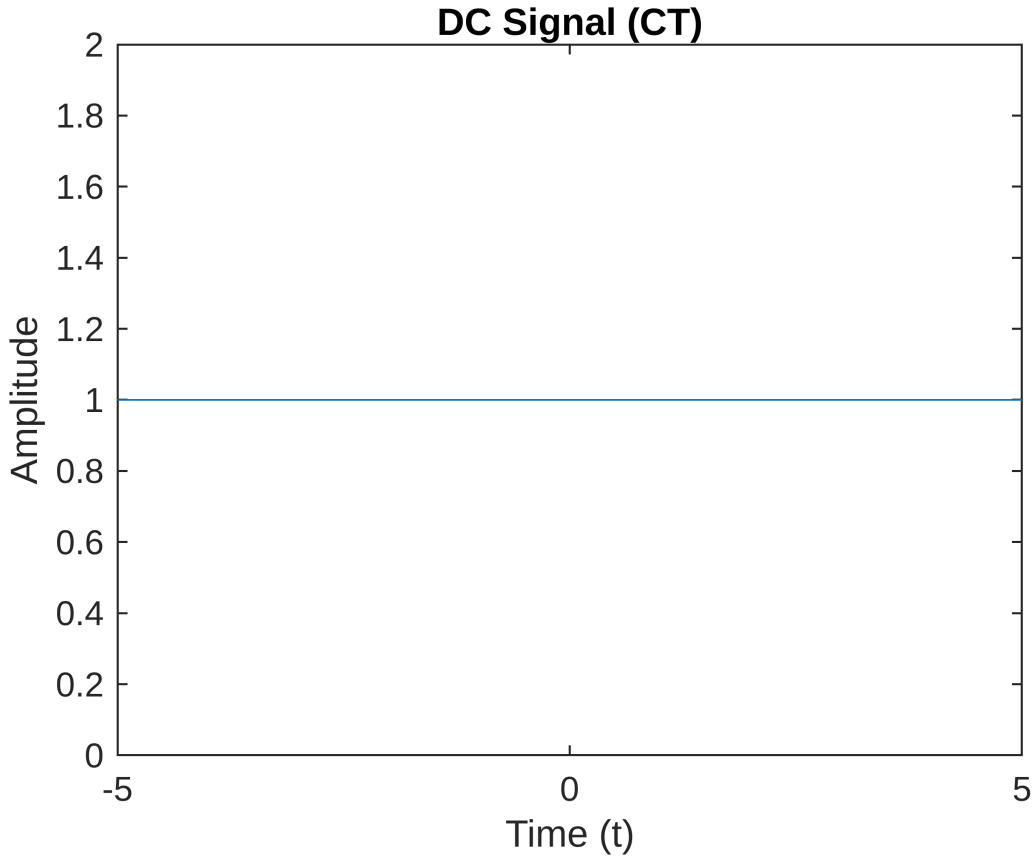

```
% 6. Program for the generation of cosine signal
% Continuous-Time (CT):
t = -5:0.01:5;    % Time vector
cos_ct = cos(2*pi*t); % Cosine in CT
figure;
plot(t, cos_ct);
title('Cosine Signal (CT)');
xlabel('Time (t)');
ylabel('Amplitude');
```



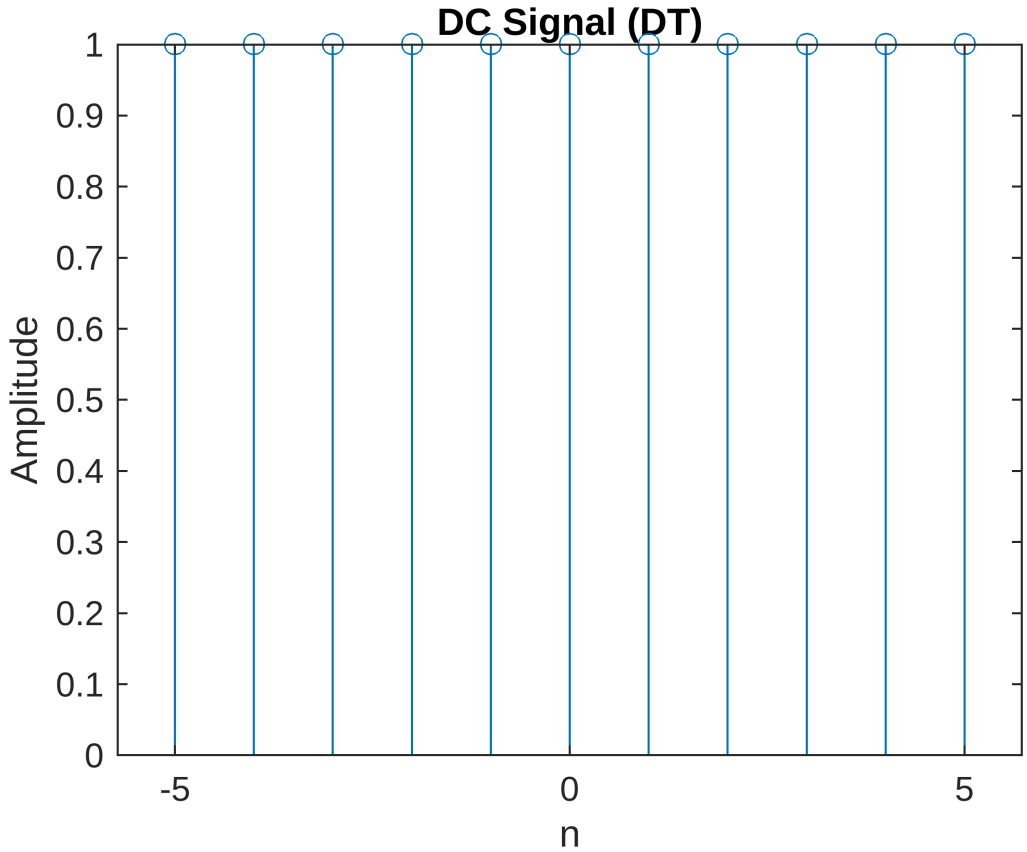

```
% Discrete-Time (DT):
n = -5:5;    % Discrete time vector
cos_dt = cos(2*pi*n/10);   % Cosine in DT
figure;
stem(n, cos_dt);
title('Cosine Signal (DT)');
xlabel('n');
ylabel('Amplitude');
```



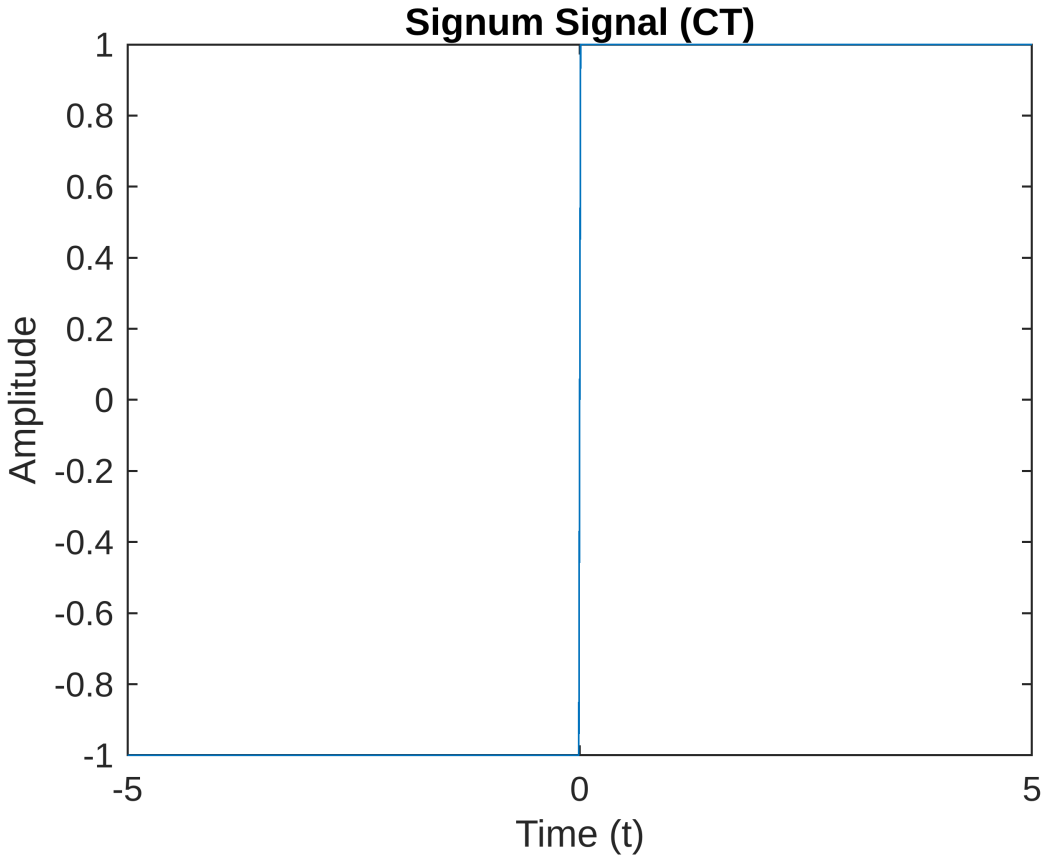

```
% 7. Program for the generation of DC signal
% Continuous-Time (CT):
t = -5:0.01:5;  % Time vector
dc_ct = ones(size(t));  % DC signal in CT
figure;
plot(t, dc_ct);
title('DC Signal (CT)');
xlabel('Time (t)');
ylabel('Amplitude');
```



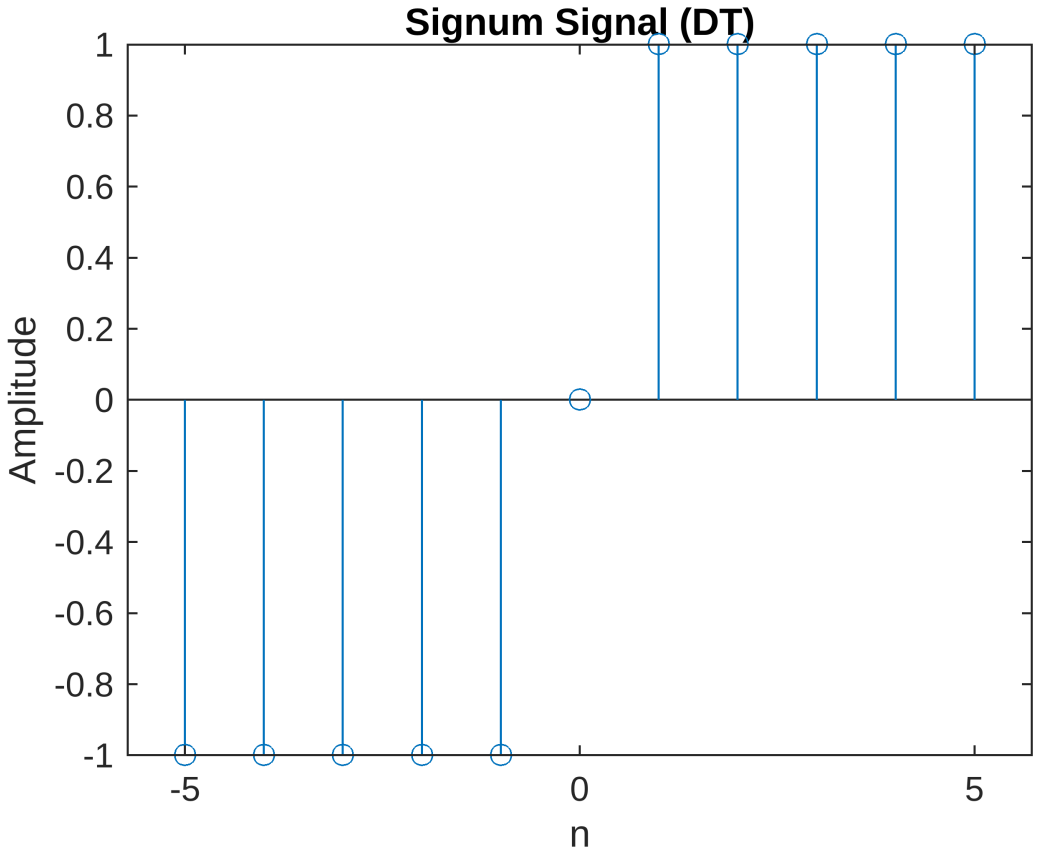

```
% Discrete-Time (DT):
n = -5:5;  % Discrete time vector
dc_dt = ones(size(n));  % DC signal in DT
figure;
stem(n, dc_dt);
title('DC Signal (DT)');
xlabel('n');
ylabel('Amplitude');
```



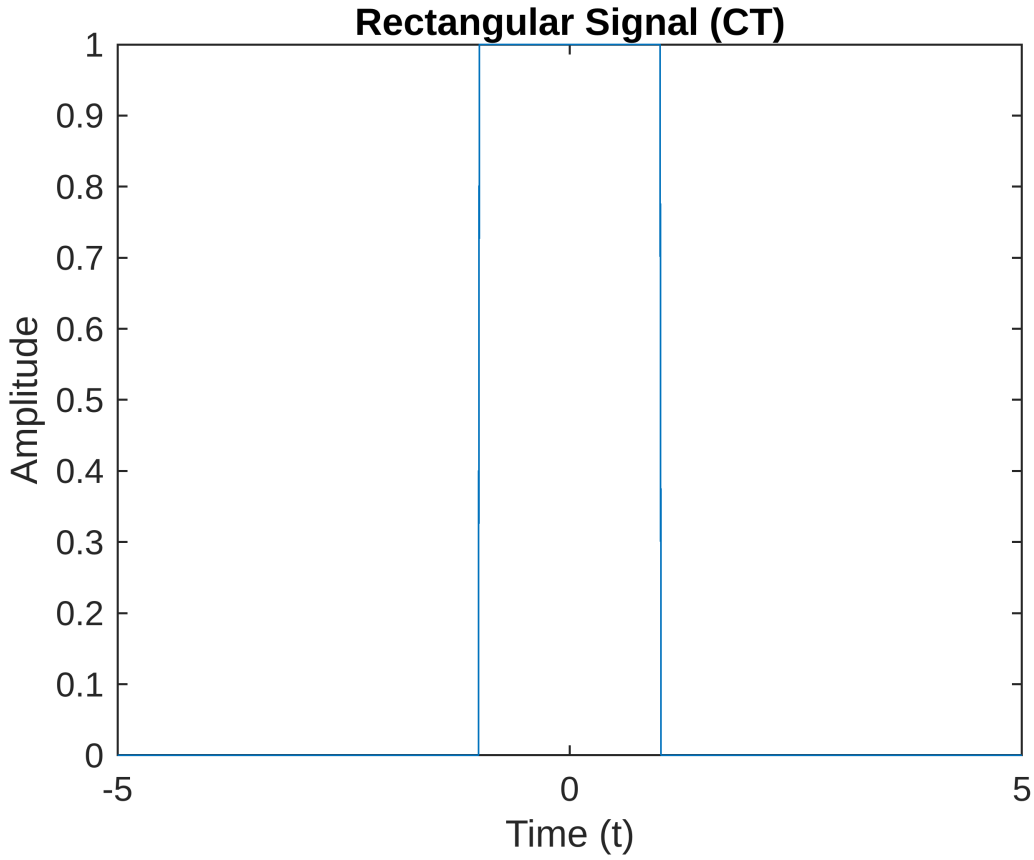

```
% 8. Program for the generation of signum signal
% Continuous-Time (CT):
t = -5:0.01:5;  % Time vector
signum_ct = sign(t);  % Signum signal in CT
figure;
plot(t, signum_ct);
title('Signum Signal (CT)');
xlabel('Time (t)');
ylabel('Amplitude');
```



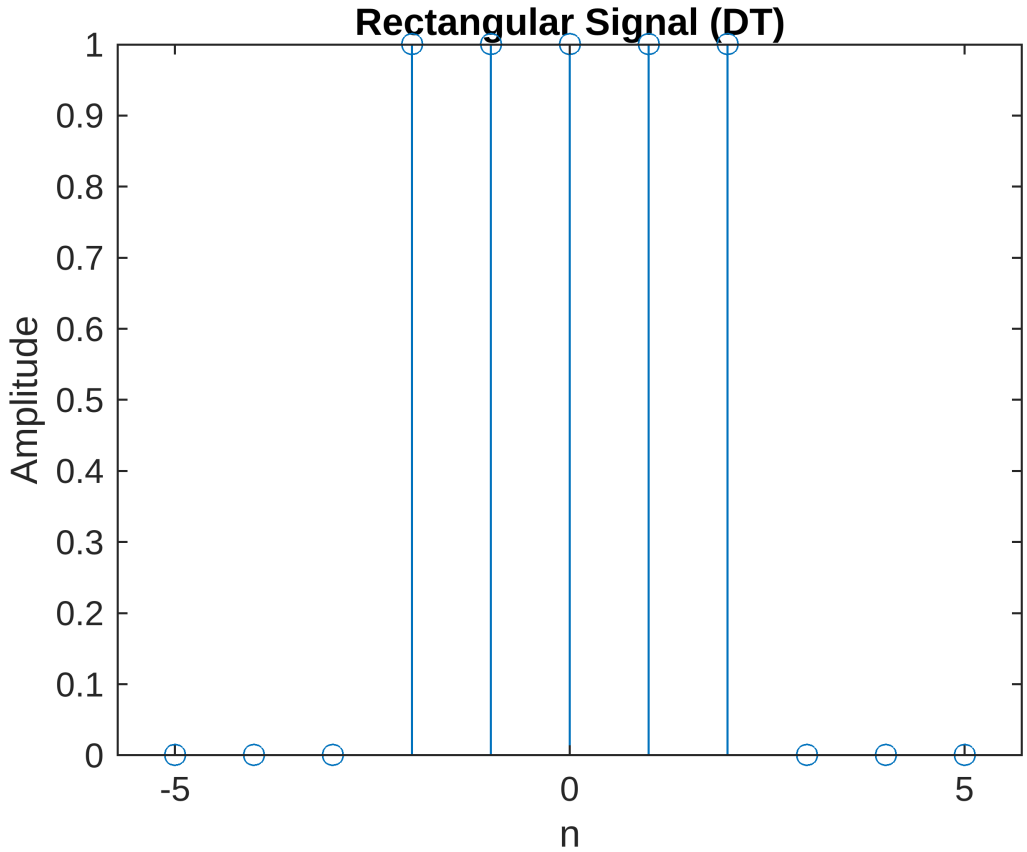

```
% Discrete-Time (DT):
n = -5:5;  % Discrete time vector
signum_dt = sign(n);  % Signum signal in DT
figure;
stem(n, signum_dt);
title('Signum Signal (DT)');
xlabel('n');
ylabel('Amplitude');
```



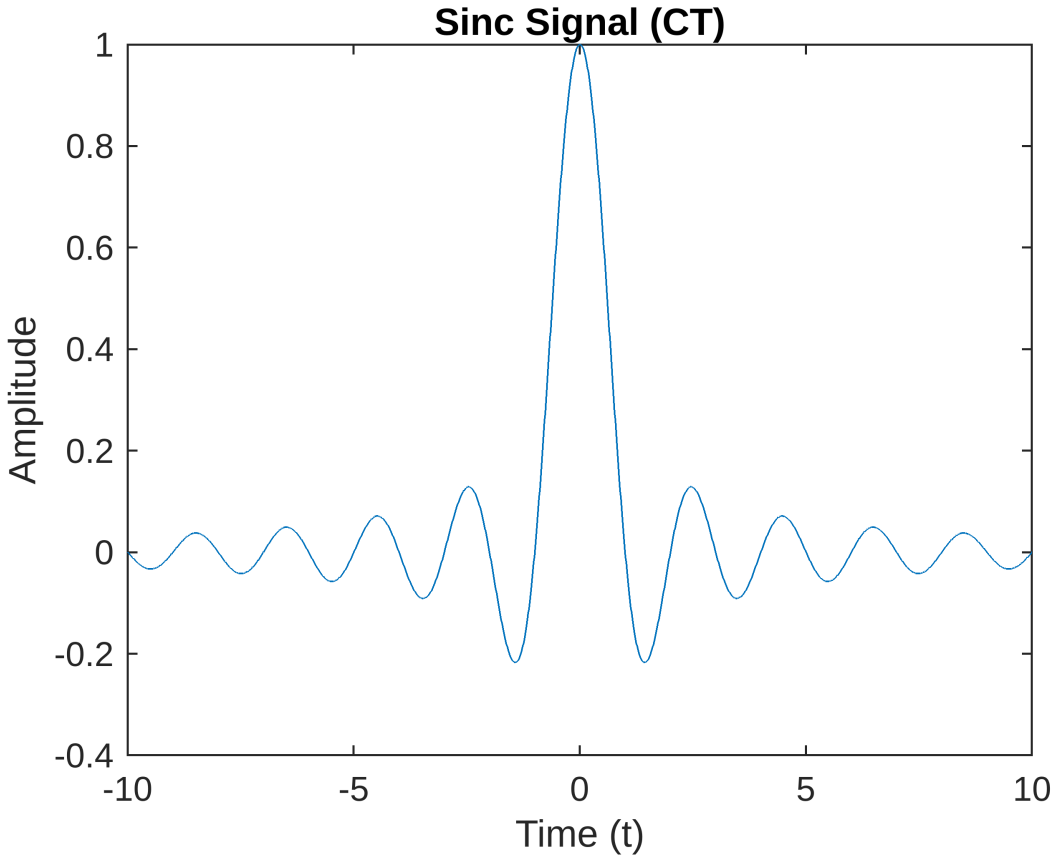

```
% 9. Program for the generation of rectangular signal
% Continuous-Time (CT):
t = -5:0.01:5;  % Time vector
rect_ct = (t >= -1 & t <= 1);  % Rectangular pulse in CT
figure;
plot(t, rect_ct);
title('Rectangular Signal (CT)');
xlabel('Time (t)');
ylabel('Amplitude');
```



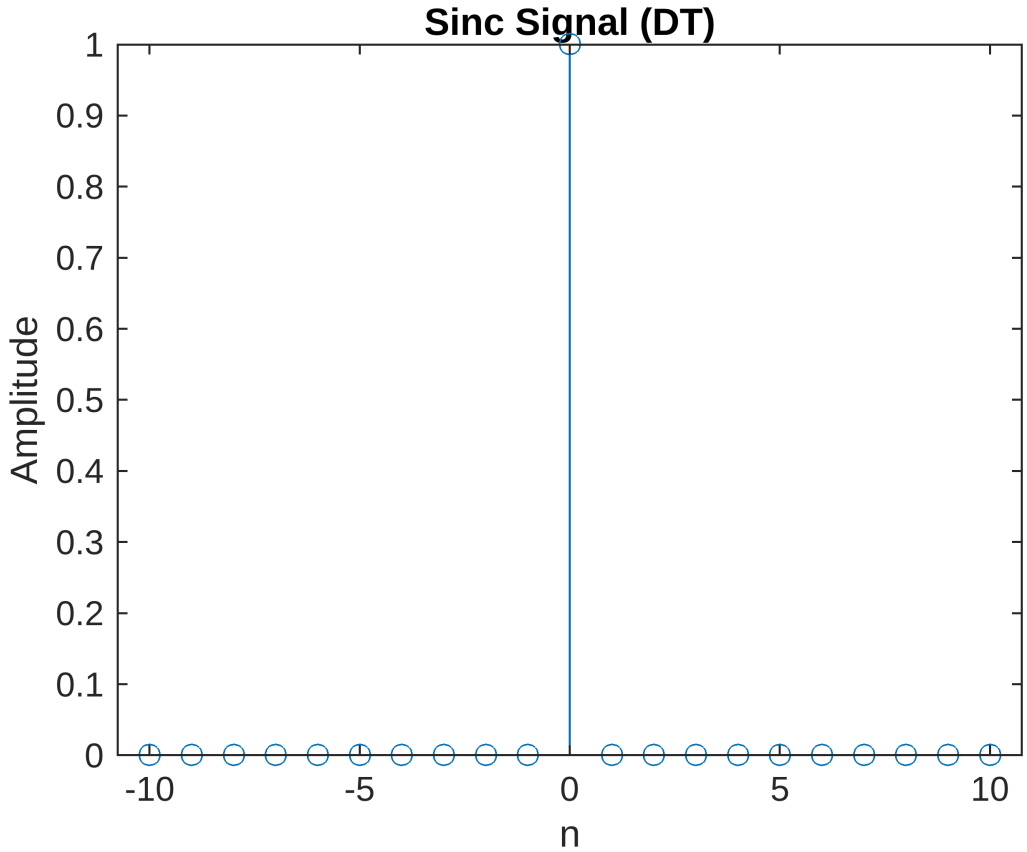

```
% Discrete-Time (DT):
n = -5:5;  % Discrete time vector
rect_dt = (n >= -2 & n <= 2);  % Rectangular pulse in DT
figure;
stem(n, rect_dt);
title('Rectangular Signal (DT)');
xlabel('n');
ylabel('Amplitude');
```

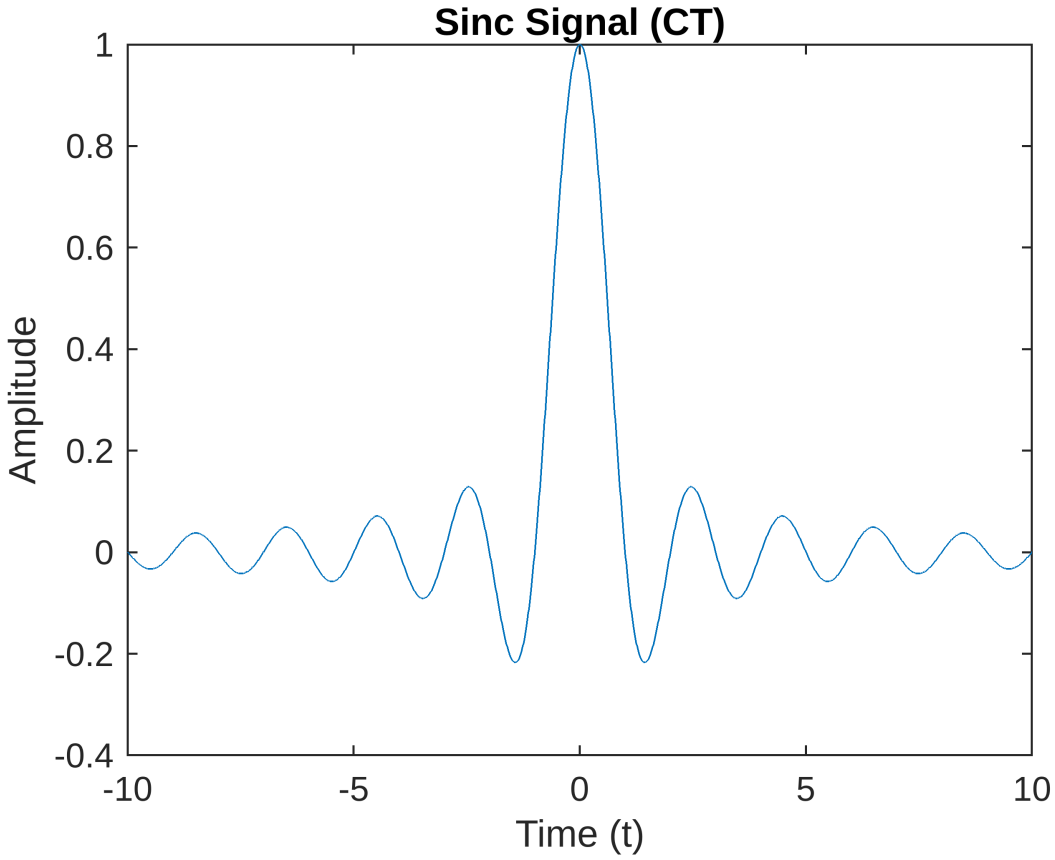



```
% 10. Program for the generation of rectangular signal
% Continuous-Time (CT):
t = -10:0.01:10;  % Time vector
sinc_ct = sinc(t);  % Sinc signal in CT
figure;
plot(t, sinc_ct);
title('Sinc Signal (CT)');
xlabel('Time (t)');
ylabel('Amplitude');
```




```
% Discrete-Time (DT):
n = -10:10;  % Discrete time vector
sinc_dt = sinc(n);  % Sinc signal in DT
figure;
stem(n, sinc_dt);
title('Sinc Signal (DT)');
xlabel('n');
ylabel('Amplitude');
```


Sinc Signal (DT)
Amplitude
n
1
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0
-10
-5
0
5
10


Sinc Signal (CT)
Amplitude
Time (t)
1
0.8
0.6
0.4
0.2
0
-0.2
-0.4
-10
-5
0
5
10